
Coding Agents for Coding Theory

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Abstract

We spent five weeks using an LLM coding agent on open problems in coding theory: finding large sets of four-letter words, such as DNA barcodes, that stay far apart in edit distance. The agent wrote the verifiers and search code; a human chose the problem and set the verification protocol. Restricting the search to codes with a prescribed symmetry, a classical technique, shrank the problem about fourfold and raised the best known code of length 6 and minimum edit distance 3 from 114 to 120 words ($E_4(6, 3) \geq 120$). The same pipeline improved twelve further lower bounds at lengths 6 to 9 and distances 3 to 6 (Table 2).

We give the failures equal space. Our own search stopped at 116 and recorded the last symmetry class as topping out at 112; a second agent session, running the same search with a better operator, found the 120. A later verdict that the method did not carry over to length 7 was wrong for the same reason, and an earlier instance (Section 5) cost three weeks. Each time, an intermediate result was written down, never rechecked, and treated as a fact that ruled out further search. Checking final outputs, as our protocol required, does not catch such errors.

1 Introduction

LLM coding agents can search large combinatorial spaces. The harder question is whether their output can be trusted. Final constructions can be verified. The intermediate conclusions an agent accumulates along the way, such as which symmetry classes have been searched and which cells are exhausted, usually are not, and those conclusions shape a project’s trajectory. This paper describes one five-week project in which agent sessions found genuine improvements to published bounds in coding theory, and also recorded wrong intermediate conclusions that hid one of those improvements for three weeks. We give the failures as much space as the results because, for anyone deciding how to supervise such an agent, the failures are more useful.

The results. The largest known quaternary code of length 6 with minimum edit distance 3, equivalently a single-edit-correcting DNA barcode set, has 120 codewords. The maintained table of bounds gives 114–176 [3], the lower value from a 2011–2012 evolutionary search [1; 2]. A recent LLM-guided search reports 113 for the same problem [7]. The 120 comes from formulating the problem as maximum independent set on a 4096-vertex confusability graph and searching over codes invariant under the Klein four-group acting on the alphabet.

Verification ran in fresh processes: reimplementations sharing no source with the search find a minimum pairwise distance of exactly 3 and zero colliding edit-balls. The confusability graph was differentially tested against the definition on all 8,386,560 vertex pairs, and reproduces the table’s exact published entries ($E_4(3, 3) = 4$, $E_4(4, 3) = 16$, $n = 5$ sweep at the published 44). Twelve further lower bounds improved (Section 4, Table 2), at $n = 6$ to 9 and d up to 6, those at $n \geq 8$ once a C orbit-graph builder made the sizes tractable. Figure 1 (Appendix B) dates each record.

2 Setup

A code $C \subseteq \Sigma^n$ over $|\Sigma| = q$ has minimum edit distance d if every pair of distinct codewords requires at least d insertions, deletions or substitutions to interconvert; $E_q(n, d)$ is the maximum size of such a code. Maximising $E_q(n, d)$ is a maximum independent set (MIS) problem on the *confusability graph*: the vertex set is Σ^n ($4^6 = 4096$ vertices at $n = 6$), with two words adjacent when their edit distance is less than d . At $n = 6$, $d = 3$ the graph has 392,358 edges, and the strong MIS solver ReduMIS [6] plateaus at 113.

What the agent did and did not do. The agent was Claude Opus 5 via Claude Code, an LLM with shell access rather than a bespoke mathematical system. It wrote the verifiers and search code and drove off-the-shelf solvers (HiGHS, CaDiCaL, SciPy/CVXPY, ReduMIS [6]). A human chose the problem, set the protocol and redirected between sessions. The agent did the implementation and interpretation, and chose methods from the protocol’s escalation order (Appendix A lists who proposed what). That protocol, which we call verifier-first, was a project instructions file loaded into every session. It required two independent verifiers with different representations and no shared code, the first written and tested on corrupted objects before any search began. It also required reproducing known published values before searching new cells, and re-verifying every saved bound in a fresh process. In the shipped version, its escalation order puts prescribed-symmetry search right after brute force and moves up a rung when the current one plateaus. The file was rewritten during the project, so we cannot tell whether that advice came before the agent first used the method. The decisive run came from a second agent session with no shared context that audited our repository (Section 3). It used its own implementation of the local search of Andrade, Resende and Werneck (ARW) [5]. Our Python operator already had the same (1,2)-swaps and plateau moves, so the difference lay in implementation details we did not isolate. Sessions did not re-read all earlier sessions. This matters for Section 5, because conclusions written in one session became invisible constraints on later ones. The verifiers are independent in representation (edit balls, two Levenshtein tables, Myers bit-vectors) but not in authorship, and the differential test against the definition on all word pairs guards their shared assumptions.

Related work. The $d = 3$ cells were recently attacked by LLM-guided search [7]. The checks we use are standard testing ideas (known-answer instances, metamorphic and differential testing), normally applied to a system’s outputs. What cost this project three weeks was an error in intermediate state and its summaries, which output checks never see.

3 What worked

Prescribed symmetry. The mathematical core is prescribed-automorphism search in the Kramer–Mesner line [4]: choose a group, require the code to be invariant under it, and search orbit representatives instead of individual words. At $n = 6$ the natural symmetry group is $S_4 \times C_2$ (the 24 permutations of the four-letter alphabet, times word reversal, giving order 48). We do not claim this is the full automorphism group of the confusability graph, which we did not compute. The group has 33 conjugacy classes of subgroups. Twenty-two are product-form (a permutation subgroup H , optionally paired with reversal). The other eleven are *diagonal*: they contain elements that pair a permutation with reversal, and cannot be written as H or $H \times C_2$. Three of the order-4 classes act freely on Σ^6 . In each, the search collapses from 4096 words to 1021 usable orbits (those whose elements are pairwise at edit distance ≥ 3), and every orbit has exactly 4 elements, so invariant code sizes are multiples of 4 and nothing lies between 116 and 120.

What our search missed. Our first sweep covered 30 of the 33 classes and gave each four seeds; the other three, missed by our enumeration, were searched later and reach 104. Two of the three free-acting classes returned 116; the third returned 112 on all four seeds. We recorded 112 as its maximum. Both parts of that recording were wrong. Re-running the same class with our operator at ten ten-minute seeds returns 116 on five of them. Then the second agent session ran its ARW implementation on the same orbit graph. It reached 116 on every seed within a minute, and 120 on two of six seeds. We then reimplemented that operator in C (`search/mis_ils.c`, about 10^3 times more iterations per second than our Python). It reproduces the 120 and finds 120s in both C_4 classes, so all three free classes contain one. Whether a seed reaches 120 is decided early: fifteen ten-minute

seeds on the free classes all returned 116, while thirty one-minute seeds on the V_4 class returned 120 four times, so more seeds beat longer ones. Those rates belong to the stronger operator. The original four-seed sweep used our own operator at 128 seconds per seed, and that operator never reaches 120 at any budget we tried. The sweep was underpowered even for 116, and its “tops out at 112” was a small sample recorded as settled fact. Nothing reached 124. A numerical, uncertified Lovász theta value on the orbit graph, about 45 orbits (180 words), is too loose to say whether a 124 exists.

The instance reduction is real. Our operator gets 111 on the raw graph at 180 core-minutes and 116 on the orbit graph at 8.5, and the stronger operator gets 120 on the orbit graph but only 114 on the raw one. ReduMIS on the same orbit graph returns 116 on each of three single-threaded ten-minute seeds, against 112–113 on the raw graph, where its seeds stopped on their own after 4.6–6.1 minutes; at the same 30 core-minutes, the C operator returns 120 on four of thirty one-minute seeds. So the reduction lifts all three operators to 116, and the step to 120 needs a stronger operator on the reduced instance, since neither alone reaches it at $n = 6$. The ranking is not stable in n . At $n = 8$ ReduMIS reaches 1124–1132 on the V_4 orbit graph against the C operator’s 1148, and at $n = 9$ it reaches 3760 against 3724 (Table 2).

Calibration cells. Verification shows that an artifact is valid. It says nothing about whether a search was strong enough for a negative result to mean anything, and most of a project’s output is negative results. We propose including in every batch one cell with an independent reference; ours is the unrestricted problem, whose exact value is open, with ReduMIS’s 113 as reference. Table 1 shows why: a sweep over 15 previously unexamined cells returned a plausible spread of values up to 100, while the same solver scored 99 on the unrestricted problem, where 113 is attainable. The batch was measuring the solver rather than the cells, so we discarded it. The check is not sufficient, since the sweep that under-searched the winning class passed its calibration cell. It detects a broken solver but not an under-budgeted one.

Table 1: Calibration across three runs, and (right) the compute-response of the repaired operator, nearly flat over 6–60 core-min; the 150 row is our earlier raw-graph local search, for reference only.

the three runs					operator vs. budget	
	run	core-min	best	calib.	budget	calib.
1	original operator	6	100	99	6 (broken op.)	99
2	repaired operator	60	112	110	6	109
3	later sweep, same op.	20	116	109	15	109
					20	109
					60	110
					150 (<i>raw-graph ILS</i>)	111

The check needs a decision rule, and an absolute threshold is the wrong one. Run 3, which produced the 116, scored 109, *below* the 110 that licensed trusting the operator in run 2, so a fixed cutoff would have discarded the run that found the 116. Normalising by compute avoids this. The right-hand columns plot the repaired operator against budget: 109 at six core-minutes and 110 at sixty, a $10\times$ increase buying one codeword. The curve has four points; two share their configuration with runs 2 and 3, so they are not independent of what is being judged. The curve is close to flat, and against it run 1’s 99 at six core-minutes is ten low. So we judge a run by its distance from the compute-response curve instead of from a fixed number. The curve costs four short runs on a known cell.

A cheaper variant needs no known answer. Conjugate subgroups give isomorphic problems, so cells within one conjugacy class must report equal optima, and any spread flags the solver. Run 1 returned 88, 87, 86, 84, 88, 83 on six isomorphic problems inside a single class.

4 Where it did not work

$n = 7$: **first a negative result, then not.** Applied unchanged to $E_4(7, 3)$, whose record is 356, the best over 30 of the 33 invariance classes was 355. (Three classes our first enumeration missed, which need three generators, were searched afterwards and reach 340.) Our notes recorded this as “the method did not transfer,” then qualified it as a statement about our search rather than about the cell, since the same operator had already under-searched the $n = 6$ winning class.

The qualification was correct. The C operator, on the same orbit graphs, returned 358 on both of its ten-minute seeds in the order-6 class that had given 355, and 359 in the reversal-only class (6844 orbits). That code consists of 164 reversal pairs and 31 palindromes. Each passed all four verifiers and a from-scratch distance check. Three further fifteen-minute seeds of the same operator on the order-6 class all gave 355, the stalling of long seeds seen at $n = 6$. Twenty-six further two-minute seeds spread 355–364, and the 364 is the value in Table 2.

Other distances. Nothing in the method is specific to $d = 3$: symbol permutations and reversal preserve edit distance, so the same 33 classes act on every conflict graph “ $\text{lev} \leq d - 1$.” We built a general- d version (a C graph builder and two independent verifiers with a 3463-check accept/reject suite) and first checked it on the table’s exact cells, which it must reproduce and must not exceed: $E_4(5, 3) = 44$, $E_4(5, 4) = 11$, $E_4(6, 5) = 8$ and $E_4(6, 6) = 4$ all came back exactly. $E_4(6, 4)$, tabled 28–32, then gave 30 in three classes, and a MaxSAT solve of one of those orbit graphs reports 30 as the optimum within that class; $E_4(7, 4)$, tabled 65–128, gave 72 in two order-8 classes. Each took under an hour including graph construction. The same pipeline then ran at $n = 7$ to 9 and at d up to 6. The winning class differs by cell (Table 2), and there was no shortcut to trying them all. A third operator, NuMVC, trailed both others everywhere.

Table 2: Every lower bound this project improved, with a symmetry class containing the record (Appendix C gives each code’s full stabiliser), the usable-orbit count of that class, and how many seeds at the stated budget reached the value (C operator unless marked ReduMIS). The four bounds above the rule were improved first; the rest came from the same pipeline extended further, with $n \geq 8$ built by a C orbit-graph builder validated edge-for-edge against the Python one. Four values are class optima as reported by exact solvers ((6, 4) by MaxSAT, the others by maximum clique of the compatibility graph); none has a checked certificate.

cell	published	new lower bound	class	orbits	seeds
$E_4(6, 3)$	114–176	120	V_4 (free)	1021	8/96 at 1 min
$E_4(7, 3)$	356–614	364	$C_3 \times C_2$	2218	2/26 at 2 min
$E_4(6, 4)$	28–32	30	diagonal, order 4, optimum	518	3/3 at 26 s
$E_4(7, 4)$	65–128	72	diagonal, order 8	1277	11/11 at 1–2 min
$E_4(8, 3)$	1132–2340	1148	V_4 (free)	16381	1/20 at 10 min
$E_4(9, 3)$	3451–9360	3760	diagonal C_4 (free)	65531	ReduMIS, 2/7 at 15–30 min
$E_4(8, 4)$	176–512	200	diagonal, order 8	6500	2/24 at 10 min
$E_4(9, 4)$	495–2048	520	diagonal, order 8	28627	4/10 at 15 min
$E_4(7, 5)$	18–24	20	V_4 (free), optimum	3568	13/13 at 0.5–1 min
$E_4(8, 5)$	38–96	44	V_4 (free)	15649	2/10 at 10 min
$E_4(9, 5)$	97–384	100	diagonal, order 8	13143	4/4 at 15 min
$E_4(8, 6)$	14–32	16	V_4 (free), optimum	12055	8/8 at 5–10 min
$E_4(9, 6)$	27–128	32	diagonal, order 8, optimum	5813	2/2 at 10 min

SAT and ILP. These end the protocol’s escalation order. For constructions they returned nothing usable. SAT reported UNKNOWN after 462 s and 389 s on instances known to be feasible (binary deletion codes at $n = 11$, from the same project), and the ILP returned unconverged incumbents on the cells here. For certificates the same tools were indispensable: drat-trim-verified UNSAT proofs, including a proof that the A_4 -invariant optimum is exactly 112, and LP duality reproducing 24 of 24 published LP relaxation values as exact rational certificates. The ranking of tools depends on direction. Heuristic local search built every construction, and formal methods produced every checked certificate. The four class optima of Table 2 have none.

5 The most expensive error

Three weeks before our first record, the 116, a sweep reported “best invariant code 112, proven optimal in the A_4 class” over what it described as all subgroups. Four defects were behind that sentence. The parameterisation expressed invariance as “ $H \leq S_4$, optionally with reversal,” reaching only the 22 product-form classes. One of the 116-word codes found later lives in a diagonal class it could not express. And a size cap skipped 15 of the 60 (subgroup, reversal) cells of that parameterisation, precisely the weakest-symmetry ones.

Inside the cap, the MILP returned unproven incumbents on cells it could not finish, and the summary dropped their status. Those last two defects are the instructive ones. Three conjugate cells, isomorphic problems that must share an optimum, were reported as 100, 92 and 8 for a quantity now known to be ≥ 120 . The results file *did* record `proven_optimal_invariant: false` for all three, and the conjugacy check of Section 3 would have flagged the spread. (The A_4 optimum really is 112; the error was “all subgroups.”) The flag was present in the data and absent from the summary, and the summary is what a human collaborator reads. None of the four raised an error. All produced plausible numbers, so a protocol that checks outputs never questioned them. The protocol had the same gap, since the ground-truth verifier did not cover this problem family until ten days after our first record, the 116. We keep the description “verifier-first” because that is what the protocol specified, not what the 116 received at the time.

Here the agent wrote conclusions faster than its supervisor re-read them, and those conclusions narrowed the search without anyone noticing. No search that trusted the recorded summary would have revisited the class that held the bound. The fix is cheap. Recheck settled conclusions on a schedule, and record unproven solver output as `unknown` rather than as a value. The protocol had no such rule until 7 September. Our own first correction, which added the diagonal classes, was itself incomplete (it enumerated 30 of the 33) and was caught only by separate review sessions.

6 Limitations

This is a single case study: one project, one problem family, one human supervisor, and no comparison across projects. The 120 came from a second agent session rather than our main search, and most later bounds from our C reimplementation of that operator ($E_4(9, 3)$ from ReduMIS). The operator comparison of Section 3, which includes ReduMIS on the orbit graphs, and the NuMVC run are the only controls we have. They show that the bounds need both the reduced instance and a strong operator, and that which operator is strongest depends on the cell. We cannot cleanly isolate how much of the result is attributable to the agent versus a conventional search pipeline using the same mathematical machinery. The key algorithmic components (prescribed symmetry, the ARW operator) predate LLMs, and the agent’s contribution was the engineering around a classical method, not a new method.

The run records sum to about 130 core-hours on one laptop. Session counts, API cost and human redirects were not logged, and intermediate conclusions were not counted as they were written, so “three times” has no denominator. What we can say is that the three negatives that pruned the search all turned out wrong when re-run at a larger budget, while others, such as the absence of a 124 at $n = 6$, held. The calibration rule has no fixed tolerance and has only been tried on the runs in Table 1, and the remedies themselves are untested. The failure modes are not specific to language models. Dropped provenance and underpowered sampling happen in any computational project, and our claim is only that here the agent wrote summaries faster than its supervisor re-read the logs. Every result is an improved lower bound, no upper bound changed, and the incumbents are unverified heuristic claims whose codeword lists were never published, so their long standing may partly reflect how few have attacked them. The logs are self-reported by the system under study, so we release them: the supplement ships the codes, all four verifiers, the defective sweep, the protocol document and every run record, and one script re-verifies every claimed bound in minutes.

7 Conclusion

The mathematics that produced the bounds is classical prescribed-automorphism search. Which operator and symmetry class worked best changed from one (n, d) to the next, and the free action fixed sizes at multiples of four without causing the escape from the incumbent. The contribution to AI methodology is the failure analysis. Three times in this project a conclusion was recorded, never rechecked, and turned out to be hiding the answer: a sweep whose summary dropped the solver’s “unproven” status, a four-seed result read as a property of a class, and a “does not transfer” verdict that was a property of the operator. A verifier-first protocol catches errors in final outputs but not in the intermediate state that decides which outputs to pursue, and we expect that gap to matter more as agents run for longer.

References

- [1] D. A. Ashlock, S. K. Houghten, J. A. Brown, J. Orth. On the synthesis of DNA error correcting codes. *BioSystems* 110(1):1–8, 2012. <https://doi.org/10.1016/j.biosystems.2012.06.005>
- [2] J. Orth. The Salmon Algorithm: A New Population Based Search Metaheuristic. MSc thesis, Brock University, St. Catharines, December 2011. Table 9.5 records $E_4(6, 3)$: previous best 108, new best 114.
- [3] S. Houghten. A Table of Bounds on Optimal Fixed-Length Quaternary Edit-Metric Codes. Brock University, <https://www.cosc.brocku.ca/~houghten/quaternaryedit.html>; the thirteen incumbents in Table 2 were read from our 2026-08-09 fetch, and the (6, 3) cell was re-checked by manual browser load on 2026-09-03 (automated fetches have returned 403 since 2026-08-26); the 2025-08-22 Wayback snapshot agrees, and the 2011 snapshot reads 111–176 at (6, 3).
- [4] E. S. Kramer and D. M. Mesner. t -designs on hypergraphs. *Discrete Mathematics* 15(3):263–296, 1976. [https://doi.org/10.1016/0012-365X\(76\)90030-3](https://doi.org/10.1016/0012-365X(76)90030-3)
- [5] D. V. Andrade, M. G. C. Resende, R. F. Werneck. Fast local search for the maximum independent set problem. *Journal of Heuristics* 18(4):525–547, 2012. <https://doi.org/10.1007/s10732-012-9196-4>
- [6] S. Lamm, P. Sanders, C. Schulz, D. Strash, R. F. Werneck. Finding near-optimal independent sets at scale. *Journal of Heuristics* 23(4):207–229, 2017. <https://doi.org/10.1007/s10732-017-9337-x>
- [7] F. Weindel and R. Heckel. LLM-Guided Search for Deletion-Correcting Codes. arXiv:2504.00613, 2025 (v2, 2026). Table 3 reports 113 at $n = 6$, 352 at $n = 7$, 1130 at $n = 8$.

A Who proposed what

Compiled from Sections 2–5 and the project notes. “Agent” means the main agent sessions; “second session” means the independent session that audited our repository and found the 120; “review sessions” are other independent adversarial-review sessions.

Decision or component	Source
Problem: open cells of the quaternary edit-metric table	Human
Protocol: verifier-first rules and escalation order	Human
Redirects between sessions	Human
Prescribed-symmetry search, a classical method [4]	Agent, within the protocol’s escalation order, which as shipped lists it second; we cannot date that entry against the agent’s first use
Verifiers, search code, orbit-graph builders, and C implementations of the ARW operator and NuMVC	Agent
Recorded conclusions, including the 112 ceiling and “did not transfer”	Agent
Its own ARW implementation [5] on the $n = 6$ orbit graph, which found the 120	Second session
Catching that our first correction enumerated only 30 of the 33 classes	Review sessions
DRAT proof that the A_4 -invariant optimum is 112	Review session
Solvers: HiGHS, CaDiCaL, drat-trim, RC2 (python-sat), SciPy/CVXPY with SCS, networkx, ReduMIS [6]	Existing tools

B Timeline

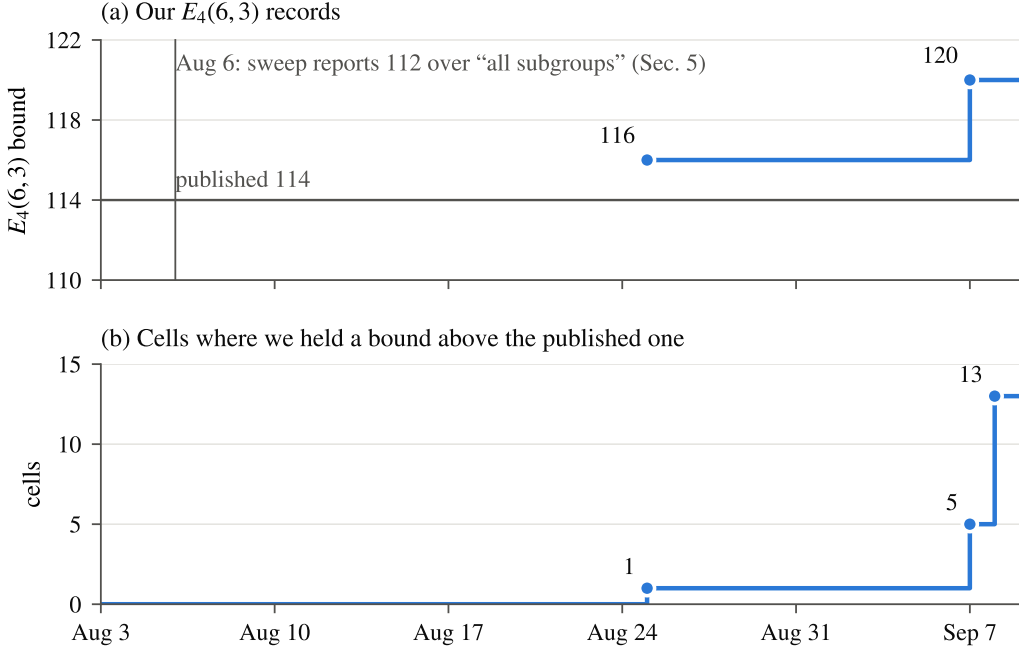


Figure 1: When each record was set, by calendar day, from the dated entries in the project notes (notes/sota.md, in the supplement). This is calendar time, not machine time: run records lack consistent timestamps and compute fields, so they cannot be placed on a machine-time axis. (a) Our $E_4(6, 3)$ records against the published 114; the vertical line marks the sweep of Section 5. (b) Cells where we held a bound above the published one, at the end of each day. Twelve of the thirteen records date from the last two days, once the ARW operator and the C orbit-graph builder were in place.

C The thirteen codes

Each list is the complete code; sha256 is over the sorted words, one per line, each line newline-terminated. Each was re-verified from the definition in a fresh process (pairwise Levenshtein distance $\geq d$). The same lists ship as plain-text files in the supplement, with the verifiers.

$E_4(6, 3) \geq 120$, **120 words**. Stabiliser in $S_4 \times C_2$: Klein four-group V_4 on symbols, no reversal. sha256 adeaa3604074f5f2b42ece623e45df96f9521650b98fb11a7182321fbc5c3576.

```
000323 001231 001302 002011 002103 003000 003222 010022 010110 011223 011311 012132
012300 013033 013201 020312 021021 021113 022030 022202 023131 023303 030013 030220
031032 031101 032233 032321 033112 033330 100200 100332 101001 101133 102122 102310
103023 103211 110213 110320 111232 112111 112333 113012 113100 120010 120123 121102
121331 122003 122221 123230 123322 130002 130130 131203 132020 132212 133121 133313
200020 200212 201121 201313 202130 203203 203331 210011 210103 211112 211330 212002
212231 213210 213323 220233 220321 221000 221222 222101 223013 223120 230122 230310
231023 231211 232200 232332 233001 233133 300003 300221 301012 301100 302232 302301
303113 303320 310030 310202 311131 311303 312220 312312 313021 320132 320300 321033
321201 322022 322110 323223 323311 330111 330333 331230 331322 332031 332102 333010
```

$E_4(7, 3) \geq 364$, **364 words**. Stabiliser in $S_4 \times C_2$: $C_3 \times C_2$, the 3-cycle (1 2 3) on symbols and reversal. sha256 dadc469672cd5dfe21dbe551ca884c81851a9cc661440f323cf7c65d895e34e6.

```
0000000 0001331 0002112 0003223 0010113 0011100 0011312 0012233 0012321 0013010
0013202 0020221 0021020 0021303 0022123 0022200 0023132 0023311 0030332 0031122
```

0031213 0032030 0032101 0033231 0033300 0100123 0100330 0101032 0103100 0103212
0110020 0111121 0111203 0112222 0112313 0113230 0120210 0121130 0122111 0122320
0123022 0130112 0130201 0131220 0131302 0132133 0133003 0133321 0200110 0200231
0201200 0201323 0202013 0210223 0210302 0211001 0211132 0212103 0212330 0213211
0220030 0221310 0222232 0222301 0223121 0223333 0230320 0231033 0232210 0233130
0233222 0300220 0300312 0302131 0302300 0303021 0310130 0311210 0311333 0312011
0313320 0320103 0320331 0321322 0322002 0322213 0323110 0323201 0330010 0331111
0331232 0332120 0333102 0333313 1000213 1000302 1001120 1002001 1003131 1010101
1011211 1012132 1012300 1013033 1020003 1020122 1020310 1021133 1022212 1023021
1023230 1031023 1032220 1032313 1033112 1100011 1101113 1101202 1102130 1103333
1110233 1110321 1111022 1111330 1112123 1112210 1113312 1120032 1121101 1122013
1122302 1123120 1123211 1130103 1130222 1131132 1131311 1132031 1133200 1200133
1201021 1203030 1203201 1203322 1210031 1211110 1213102 1213220 1220200 1220323
1221332 1222022 1222131 1223113 1230002 1230111 1231212 1231303 1232100 1232321
1233233 1233310 1300121 1301223 1302102 1302311 1310332 1311103 1312030 1312221
1313001 1313213 1320020 1321112 1321231 1322333 1323222 1323300 1330230 1331000
1332012 1332203 1333123 1333331 2000321 2000321 2001212 2002230 2003002 2011223
2012031 2013121 2013330 2020202 2021011 2022322 2023100 2023213 2030001 2030120
2030233 2031032 2031310 2032211 2033323 2100232 2102331 2103122 2103203 2110310
2111112 2111231 2112000 2113023 2113301 2120113 2121002 2121321 2122201 2123010
2123332 2130030 2131100 2131333 2132223 2132312 2133111 2200022 2201111 2202221
2202303 2203210 2210201 2210333 2211300 2212122 2212213 2213012 2220132 2220311
2221123 2222033 2222110 2223231 2223320 2230013 2231230 2231322 2232202 2233021
2233103 2300211 2301010 2301133 2301302 2302032 2310003 2310222 2311120 2311311
2312101 2312323 2313132 2313200 2320012 2321203 2321330 2322220 2330131 2330300
2331221 2332113 2333033 2333212 3000132 3000201 3001003 3002323 3003310 3010002
3010230 3010311 3011131 3012013 3012120 3013322 3021110 3021232 3022331 3023012
3030303 3031200 3031321 3032022 3033133 3100322 3102020 3102103 3102211 3103013
3110100 3110212 3111011 3111323 3112332 3113221 3120001 3120333 3121213 3121300
3122122 3122230 3123131 3123202 3130023 3132110 3132301 3133330 3200313 3201233
3201301 3203112 3210010 3211222 3212111 3212200 3213123 3213331 3220120 3221031
3221102 3222223 3222312 3223000 3230221 3231020 3231113 3232003 3232132 3233302
3300033 3301320 3302222 3303101 3303332 3310021 3311032 3311201 3312133 3312310
3313303 3320111 3320302 3321023 3322100 3323233 3323321 3330122 3330213 3331130
3331312 3332231 3333011 3333220

$E_4(8, 3) \geq 1148$, **1148 words.** Stabiliser in $S_4 \times C_2$: Klein four-group V_4 on symbols, no reversal.
sha256 e98fe9d121189c38208634f4b8978be329f6ae0f2d2bedd97105bb87c57fb8.

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$E_4(9, 3) \geq 3760$, **3760 words**. Stabiliser in $S_4 \times C_2$: the diagonal C_4 generated by $((0\ 2\ 1\ 3), \text{rev})$;
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333021130 333022321 333030010 333030333 333031003 333031222 333033120 333100103
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333113210 333120021 333120232 333122010 333123303 333130213 333131123 333132230
333132311 333133102 333200110 333200332 333201211 333201323 333202202 333203131
333210203 333211010 333211331 333212312 333213300 333222100 333223001 333223113
333223220 333230002 333230320 333232013 333232122 333233232 333300011 333301203
333301312 333302210 333310310 333311001 333311220 333312130 333313022 333320003
333321333 333322111 333323202 333330101 333331110 333332221 333332303 333333000

$E_4(6, 4) \geq 30$, **30 words.** Stabiliser in $S_4 \times C_2$: {id, (0 1)(2 3), ((2 3), rev), ((0 1), rev)} (diagonal, order 4). sha256 fc6fb24268384a77f148646c7760bb3049b57258d28386310f392f89675a270f.

000333 001011 003122 013210 020202 021130 032231 033003 102301 110100 111222 112033
122112 123320 130021 131313 200220 201103 212121 220311 222000 223233 230132 303030
310012 311331 321023 331200 332322 333111

$E_4(7, 4) \geq 72$, **72 words**. Stabiliser in $S_4 \times C_2$: the order-8 diagonal group generated by $((01), \text{rev})$, $((23), \text{rev})$ and $(02)(13)$. sha256
85cb1ceb68de50ccc1566cdddc8240d28b328cbb2b688e8cc0520bc74aeb391.

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0000022 0001210 0012221 0020300 0022132 0033333 0110212 0120033 0131000 0203223
0210130 0221103 0222011 0233112 0303302 0311313 0323231 0330220 1001303 1020111
1031122 1103330 1110301 1111133 1122222 1131211 1133023 1200202 1212213 1221331
1232320 1301021 1312332 1322003 1330012 1333100 2000233 2003321 2011330 2021001
2032312 2101013 2112002 2121120 2133131 2200310 2202122 2211111 2222200 2223032
2230003 2302211 2313222 2332030 3003113 3010102 3022020 3030031 3100221 3111322
3112230 3123203 3130110 3202333 3213300 3223121 3300000 3311201 3313033 3321112
3332123 3333311
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$E_4(8, 4) \geq 200$, **200 words**. Stabiliser in $S_4 \times C_2$: the order-8 diagonal group generated by $((01), \text{rev})$, $((23), \text{rev})$ and $(02)(13)$. sha256
2083125e16976bdc2c73a6de79fef82e798668ebb92c84ac32b9c3a4c148d251.

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00003213 00012232 00022100 00032021 00103332 00111013 00130122 00133000 00200300
00210211 00220023 00231233 00311200 00321032 00333311 01002333 01023010 01031212
01113233 01120202 01132110 01211310 01221122 01230032 01233201 01300210 01310301
01320133 01331023 02000120 02011030 02021323 02101021 02123111 02130313 02212012
02223330 02232220 02302003 03001131 03020312 03030223 03100030 03102222 03111120
03222231 03233322 03312113 03313002 03332330 10002322 10023001 10031313 10113222
10120303 10132101 10201210 10211301 10220132 10231022 10300201 10321123 10322310
10330033 11000102 11012223 11021033 11022111 11103323 11112302 11123130 11133011
11200311 11222200 11230123 11301300 11311211 11320322 11331132 12000031 12011121
12013333 12110020 12121332 12131203 12202113 12203002 12223221 12322233 12333320
13010130 13021202 13032000 13100121 13111031 13130232 13213112 13303103 13323331
13332221 20001112 20010002 20030230 20120221 20203101 20222302 20233212 20301333
20312131 20323203 21000013 21011100 21110112 21130331 21131220 21202130 21212001
21223313 21320000 21322212 21333302 22002201 22013011 22022122 22032033 22103210
22111133 22133022 22200322 22210203 22221031 22230010 22311222 22312300 22321110
22333231 23003300 23011023 23012210 23033132 23102311 23113201 23122032 23132123
23201232 23213030 23220111 23302020 23310332 23331011 30001003 30020331 30021220
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31101113 31110003 31121321 31203020 31210222 31232312 31312010 31322303 31333213
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32213131 32220100 32302121 32310323 32331000 33000022 33012301 33022133 33102100
33113310 33123122 33133033 33200333 33203211 33222320 33230001 33301312 33311233
33321101 33330120
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$E_4(9, 4) \geq 520$, **520 words**. Stabiliser in $S_4 \times C_2$: the order-8 diagonal group generated by $((01), \text{rev})$, $((23), \text{rev})$ and $(02)(13)$. sha256
f48053ee681fc5791c5a4f367cecd8c5bbd3d32c45dd1ffe10c194e5f5746569.

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000000211 000011031 000023301 000030333 000101323 000112300 000121012 000202222
000203000 000222130 000331122 000333230 001100102 001133203 001220020 001232011
001310231 001322223 002003110 002013022 002111111 002112232 002203321 002211333
002232123 002300013 002311000 002320202 003001332 003021221 003120110 003123033
003213112 003300320 003333311 010003121 010022033 010103032 010112221 010210230
010300001 010302113 010323222 011010110 011031200 011110322 011113101 011200300
011201221 011222112 011233133 011302202 012011132 012033313 012100212 012121031
012223023 012232032 012322120 012333201 013010023 013101122 013113330 013132323
013203210 013220103 013321111 013331302 020001002 020102312 020121220 020230112
020311101 020332023 021003322 021032303 021103113 021111202 021130021 021312130
021320012 021333000 022000223 022012001 022021311 022120133 022133120 022200101
022211210 022222221 022310303 022331113 023003231 023013010 023202333 023211032
023222000 023301203 023312211 023330131 030002132 030031223 030033100 030133321
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031231002 031313013 031332310 032002200 032022113 032030302 032103011 032220122
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033233303 033300112 100001233 100002010 100101001 100120311 100213313 100310330
100311211 100322022 100333003 101003330 101012123 101112030 101133122 101211110
101213002 101232333 101301321 102002221 102010033 102023232 102101132 102220213
102230000 102312301 102331012 103011303 103030120 103100023 103122202 103222310
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103233031 103323123 103332132 110011013 110022312 110201320 110233332 110323100
110331131 111003211 111010232 111030103 111100120 111111300 111121222 111132210
111220033 111222321 111312111 111313333 111333021 112031001 112032122 112110223
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123133002 123200230 123223322 123331033 130000313 130012002 130021130 130112233
130123212 130203021 130222111 130231103 131013203 131030331 131110113 131200010
131223132 131321003 132102101 132112320 132203300 132210312 132221020 132300123
132313222 132333111 133022031 133031022 133103110 133111332 133130200 133201212
133220002 133300301 133311010 133333330 200000003 200022323 200033032 200113331
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202133323 202223220 202303002 202320130 203102230 203111222 203130312 203210121
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210212120 210220022 210321233 211011121 211022213 211032031 211122330 211201130
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212220310 212311103 213003112 213013220 213033301 213110132 213131231 213202023
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221223110 221301211 221302332 222000312 222020000 222021222 222111012 222113300
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230111023 230211131 230233310 230303213 230322030 231002321 231021032 231103333
231113120 231232201 231310101 231323300 231331112 232032012 232101000 232120331
232122223 232200211 232221303 232321210 232330003 233000330 233011311 233022122
233023003 233120020 233213022 233232332 233331323 233332100 300033221 300100030
300123120 300133302 300202303 300213101 300221231 300310021 301001100 301022012
301113211 301230322 301303031 301311220 301331133 302001023 302020320 302102331
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303200012 303300233 303302110 303331201 310003202 310021122 310032130 310111333
310122301 310131000 310320323 310330102 311002220 311023030 311111112 311122123
311133232 311200213 311213200 311312022 311321332 311333110 312000333 312013321
312021203 312203312 312222131 312230220 312301030 312330011 313001310 313022232
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322133033 322220232 322223011 322302133 322323223 323010111 323031220 323033332
323123103 323221112 323230301 323311300 323330212 330000022 330033013 330120221
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331122000 331130012 331221101 331222222 331320311 331330223 332011110 332023102
332101322 332113313 332200130 332233231 333000103 333002211 333111203 333130333
333131111 333212321 333221033 333232010 333303000 333310032 333322302 333333122

$E_4(7, 5) \geq 20$, **20 words.** Stabiliser in $S_4 \times C_2$: Klein four-group V_4 on symbols, no reversal.
sha256 74f19f275251e84fc6cde1a1dc58efedefb314f53d0c830e5021537f7438a553.

0003332 0031211 0110012 0222202 0311320 1001103 1112223 1120300 1200231 1333313
2000020 2133102 2213033 2221110 2332230 3022013 3111131 3223321 3302122 3330001

$E_4(8, 5) \geq 44$, **44 words.** Stabiliser in $S_4 \times C_2$: Klein four-group V_4 on symbols, no reversal.
sha256 b777627b5733905edf22910b860ee33ed56227f5f2e18ba73d090b27b20fdc99.

00033021 00202003 00221212 01000120 01231133 01312201 02111110 02132322 02223331
03012032 03323000 10111031 10203310 10320022 11122130 11313112 11330303 12103123
12232111 13000001 13023233 13332220 20001113 20310100 20333332 21101222 21230210
22003030 22020221 22211203 23013311 23130023 23222302 30010333 30321301 31110002
31201011 31222223 32021132 32102200 32333213 33112121 33131330 33300312

$E_4(9, 5) \geq 100$, **100 words.** Stabiliser in $S_4 \times C_2$: the order-8 diagonal group generated by $((01), \text{rev})$, $((23), \text{rev})$ and $(02)(13)$. sha256
f0647e8c174766334d8c526b705cc53cc77a5d21fb6a959e1f7ec85dd51eea4a.

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000000131 000013202 000302000 000321221 002211031 002322332 003310323 010230022
011001033 011331302 012232201 013010210 013120301 020022230 022023111 022203330
023112013 023330030 030111111 030123333 031002211 031211223 032133012 033200113
033221002 100110122 100220213 101321133 102031210 102101301 103323310 111102313
111111020 111213111 111230330 112201232 113233223 113300120 120113300 120300332
121000000 121032222 122311002 122330113 123022103 131133321 132003102 132221121
133132000 133312221 200021112 200201333 201112212 201330231 202200012 210311230
211003220 211022331 212301111 212333333 213033001 213220033 220033213 220100110
221132101 222103003 222120222 222222313 222231020 230010023 231232032 231302123
232012200 233113120 233223211 300112331 300133220 301200321 302122110 302331122
303210000 303222222 310003303 310221320 311130003 311310222 313311103 320213032
320323123 321101132 322002031 322332300 323103311 330023010 331011001 331122302
333012112 333031333 333320131 333333202

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$E_4(8,6) \geq 16$, **16 words**. Stabiliser in $S_4 \times C_2$: $V_4 \times \langle \text{rev} \rangle$ (order 8). sha256
773280cf9dc013ce9337801d25485c142dfa0ca8d2f17d10faa71c9f575d903f.

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00033000 01131332 02100120 03332221 10020223 11122111 12223330 13011031 20322302
21110003 22211222 23313110 30001112 31233213 32202001 33300333

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$E_4(9,6) \geq 32$, **32 words**. Stabiliser in $S_4 \times C_2$: the order-8 diagonal group generated by $((01), \text{rev})$, $((23), \text{rev})$ and $(02)(13)$. sha256
f034a20a9dac757148b2ea2653eca03b60a5f233aa60dde872f667e20c8ef99.

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000000322 000111000 001122112 001333333 021031021 022002233 030231113 033301202
110033003 110222222 111000111 111111233 121320002 122210313 130120130 133113322
200220011 203213203 211123020 212013331 222222100 222333222 223111111 223300330
300032131 303102220 311331100 312302312 332000000 332211221 333222333 333333011

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